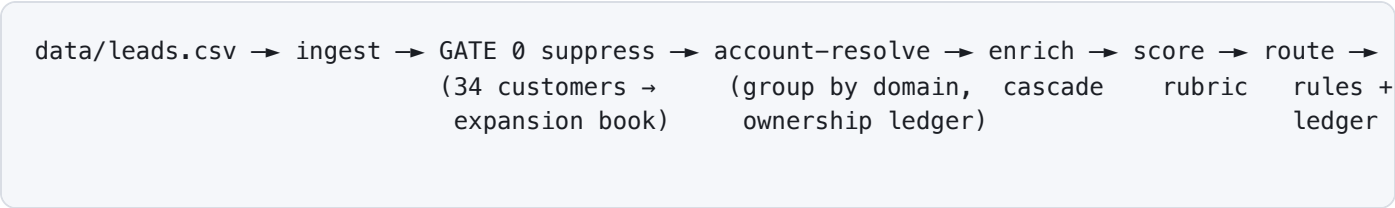


Architecture — One Page

30 mock leads → suppress → enrich → score → route → enable. **Deterministic pipeline (0-100-0 core) with two gated language leaves** — the LLM appears only where language is the deliverable or the world is genuinely ambiguous; code decides everything a metric tracks.



Step by step

#	Stage	What happens	Executor / API	Tier	Decision source
1	Ingest	Parse 30 leads; identity fields (name/email/domain/title) never mutated	lib/csv.ts	SYMBOLIC	—
2	Gate 0 — suppress	Registrable-domain match vs. known-customer list → expansion book; never enriched on acquisition spend	lib/suppression.ts + customer CSV	SYMBOLIC	customer list (set membership)
3	Account resolution	Group leads → accounts by registrable domain; account.score = MAX(leads); ownership-ledger check	lib/domain.ts, lib/ownership.ts	SYMBOLIC	ownership ledger
4	Enrich — cascade	BitScale (0.9) → nRev (0.85) → DNS MX/RDAP (0.75, free/keyless) → symbolic (0.6) → LLM residue (≤0.6 code-clamped; classifies, never scores; fires only when all sources miss or confidence <0.5)	connections/*, lib/enrich.ts	SYMBOLIC + NEURAL (bounded)	config/enrichment.js

5	Score	4-dimension rubric, 0–100; hot ≥75 · warm 50–74 · cold <50; domain evidence outranks LLM labels	lib/scoring.ts	SYMBOLIC (feeds tracked metrics)	config/icp-rubric.js
6	Route	Owned → owner (never reassigned); unowned → vertical→rep rules; tier → P1/P2/P3; unmatched → Rep C, flagged	lib/routing.ts	SYMBOLIC	config/routing-rules.json
7	Enable — battle cards	Hot/warm accounts only; 7-section card; approved proof points only; Langfuse groundedness eval (uncited claim = failed card)	tools/battle-card.ts + prompts/battle-card.md	NEURAL → HUMAN	grounding corpus + blocklist/allowlist
8	Decision brief	Code computes deltas vs. prior run, enrichment trust score, projections, 4Ws recommendations; LLM optionally narrates the computed numbers — never computes	tools/decision-brief.ts + memory/runs/	SYMBOLIC (0-100-5)	prior-run snapshot

Architecture note — tools/APIs, hosting, data model, why

Concern	Choice	Why
Hosting & orchestration	Trigger.dev (TypeScript tasks, cloud-deployed); local <code>npm run pipeline</code> needs zero keys	Per-stage retries, structured logs, shareable run link — "it runs" is a live artifact, and pipeline-health instrumentation comes free
Enrichment APIs	Bitscale (0.9) → nRev (0.85) → DNS MX + RDAP (0.75, free/keyless) → symbolic → LLM residue	Brief asks for a free/public method — DNS/RDAP return live truth even on mock domains. Vendors are env-gated and probe-first (verify response shape on 1 lead before spending on 30); a fallback is never dressed as vendor data

LLM	Claude Haiku (Anthropic API) at exactly 3 touchpoints: enrichment residue, battle cards, brief narration	Cards and narration are language deliverables; everything a metric tracks stays in cheaper, testable code
LLM observability	Langfuse — traces every call (prompt, tokens, cost, latency) + groundedness eval per battle card	An uncited claim fails the card before a rep sees it; LLM spend is a dashboard number, not a surprise
Delivery & alerting	Slack (via Trigger.dev task — production rail, not in the demo run): P1 hot-lead alert + battle card to the routed rep; run summary + threshold alerts to ops; Part 2 reuses the rail for CSM expansion alerts	Reps live in Slack, not in CSV files — the routed lead has to land where the SLA clock starts
Decision logic	All in <code>config/</code> (<code>icp-rubric.json</code> , <code>routing-rules.json</code> , <code>enrichment.json</code>)	Changing the ICP or territories is a config edit, never a code edit; the rubric carries its evidence citations inline
Outputs	<code>routed_leads.csv/.json</code> · <code>battle_cards/*.md</code> · <code>decision_brief.md</code> + <code>brief.json</code>	CSV for the brief's deliverable, JSON for machines, cards for reps, brief for the operator

Data model

One `Lead` accretes (source fields never mutated), grouped into `Account`:

```
{ lead → enrichment{source,confidence} → score{total,breakdown,tier} → routing{rep,priority,ow
```

The same account shape serves both bowtie sides — Part 2 re-points enrich→score→route at the expansion book.

Measured neural authority (live run)

LLM-touched enrichment: **26/30 leads** (mock domains — vendors/public sources miss; rare on real data). Neural authority over tracked outputs: **0/30 tiers, 0/30 routes**; one within-tier score nudge where the domain carried no evidence; in its only conflict with a domain keyword, the LLM lost. **Coverage ≠ authority.**

Patterns deliberately not used: ReAct · plan-and-execute · orchestrator-worker · ToT · ensemble · reflexion — the full flowchart is drawable before runtime, so model-driven control flow adds only cost and failure modes. Learning loop (rep accept/decline ledger retuning weights) is a documented extension.

